CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

31. Match **List I** with **List II**:

	List I (Complexes)		**List II (Hybridisation)**
A.	$[Ni(CO)_4]$	I.	sp^3
B.	$[Cu(NH_3)_4]^{2+}$	II.	dsp^2
C.	$[Fe(NH_3)_6]^{2+}$	III.	sp^3d^2
D.	$[Fe(H_2O)_6]^{2+}$	IV.	d^2sp^3

(1) A-II, B-I, C-IV, D-III (2) A-I, B-II, C-IV, D-III

(3) A-I, B-II, C-III, D-IV (4) A-II, B-I, C-III, D-IV

Answer (2)

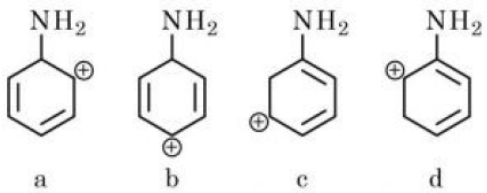
Sol. $Ni(CO)_4 \longrightarrow sp^3$

$[Cu(NH_3)_4]^{2+} \longrightarrow dsp^2$

$[Fe(NH_3)_6]^{2+} \longrightarrow d^2sp^3$

$[Fe(H_2O)_6]^{2+} \longrightarrow sp^3d^2$

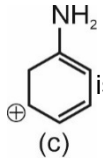
32. The most stable carbocation for the following is:



(1) a (2) b

(3) d (4) c

Answer (3)

Sol.  is the most stable carbocation because its resonance goes up to nitrogen atom.

(c)

33. Which of the following reaction is correct?

(1) $4LiNO_3 \xrightarrow{\Delta} 2Li_2O + 2N_2O_4 + O_2$

(2) $2LiNO_3 \longrightarrow 2Li + 2NO_2 + O_2$

(3) $2LiNO_3 \xrightarrow{\Delta} 2NaNO_2 + O_2$

(4) $4LiNO_3 \xrightarrow{\Delta} 2Li_2O + 4NO_2 + O_2$

Answer (4)

Sol. $4LiNO_3 \xrightarrow{Heat} 2Li_2O + 4NO_2 + O_2$

$LiNO_3$ on heating produces Li_2O, NO_2 and O_2

34. The wave function (Ψ) of 2s is given by

$$\Psi_{2s} = \frac{1}{2\sqrt{2\pi}}\left(\frac{1}{a_0}\right)^{1/2}\left(2-\frac{r}{a_0}\right)e^{-r/2a_0}$$

At r = r_0, radial node is formed. Thus, r_0 in terms of a_0

(1) $r_0 = 2a_0$ (2) $r_0 = 4a_0$

(3) $r_0 = a_0$ (4) $r_0 = \frac{a_0}{2}$

Answer (1)

Sol. For radial node $\Psi_{2s} = 0$

$\therefore r = 2a_0$

35. 1 L, 0.02 M solution of $[Co(NH_3)_5SO_4]$ Br is mixed with 1 L, 0.02 M solution of $[Co(NH_3)_5Br]SO_4$. The resulting solution is divided into two equal parts (X) and treated with excess of $AgNO_3$ solution and $BaCl_2$ solution respectively as shown below:

1 L Solution (X) + $AgNO_3$ solution (excess) $\rightarrow$ Y

1 L Solution (X) + $BaCl_2$ solution (excess) $\rightarrow$ Z

The number of moles of Y and Z respectively are

(1) 0.02, 0.02 (2) 0.01, 0.01

(3) 0.02, 0.01 (4) 0.01, 0.02

Answer (2)

Sol. On misxing both $\left[Co(NH_5)_5SO_4\right]Br$

and $\left[Co(NH_3)_5Br\right]SO_4$ becomes 0.01 molar.

$\therefore$ moles of y and z formed will also be 0.01 both.

36. Bond dissociation energy of "E-H" bond of the "H_2E" hydrides of group 16 elements (given below), follows order.

A. O B. S

C. Se D. Te

Choose the correct from the options given below:

(1) A > B > C > D (2) B > A > C > D

(3) A > B > D > C (4) D > C > B > A

Answer (1)

Sol. The correct order of bond strength is

$H_2O > H_2S > H_2Se > H_2Te$

37. Chlorides of which metal are soluble in organic solvents?

(1) Be (2) Mg

(3) K (4) Ca

Answer (1)

Sol. $BeCl_2$ is a covalent molecule

So, it is soluble in organic solvents, rest are ionic compounds.

38. Match List I with List II.

	List I (Mixture)		**List II (Separation Technique)**
A.	$CHCl_3 + C_6H_5NH_2$	I.	Steam distillation
B.	$C_6H_{14} + C_5H_{12}$	II.	Differential extraction
C.	$C_6H_5NH_2 + H_2O$	III.	Distillation
D.	Organic compound in H_2O	IV.	Fractional distillation

(1) A - III, B - I, C - IV, D - II

(2) A - IV, B - I, C - III, D - II

(3) A - III, B - IV, C - I, D - II

(4) A - II, B - I, C - III, D - IV

Answer (3)

Sol.

Mixture		Separation technique
(A) $CHCl_3 + C_6H_5NH_2$	$\rightarrow$	Distillation
(B) $C_6H_4 + C_5H_{12}$	$\rightarrow$	Fractional distillation
(C) $C_6H_5NH_2 + H_2O$	$\rightarrow$	Steam distillation
(D) Organic compound in H_2O	$\rightarrow$	Differential extraction

39. Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: Antihistamines do not affect the secretion of acid in stomach.

Reason R: Antiallergic and antacid drugs work on different receptors.

In the light of the above statements, Choose the correct answer from the options given below.

(1) A is false but R is true

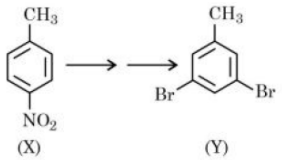
(2) Both A and R are true and R is the correct explanation of A

(3) Both A and R are true but R is not the correct explanation of A

(4) A is true but R is false

Answer (2)

Sol. Antihistamines not affect the secretion of acid in stomach, the reason is that antiallergic and antacid drugs work on different receptors.

40. 

(X) (Y)

In the above conversion of compound (X) to product (Y), the sequence of reagents to be used will be

(1) (i) Fe, H^+ (ii) Br_2(aq) (iii) HNO_2 (iv) CuBr

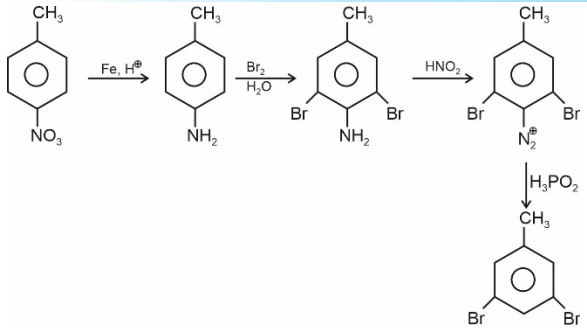
(2) (i) Br_2(aq) (ii) $LiAlH_4$ (iii) H_3O^+

(3) (i) Fe, H^+ (ii) Br_2(aq) (iii) HNO_2 (iv) H_3PO_2

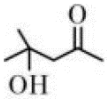
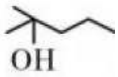
(4) (i) Br_2, Fe (ii) Fe, H^+ (iii) $LiAlH_4$

Answer (3)

Sol.



41. Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A :  can be easily reduced using Zn-Hg/HCl to  .

Reason R : Zn-Hg/HCl is used to reduce carbonyl group to $-CH_2-$ group.

In the light of the above statements, choose the **correct** answer from the options given below :

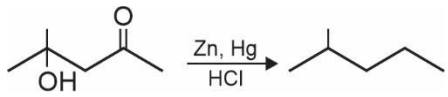
(1) Both A and R are true and R is the correct explanation of A

(2) A is true but R is false

(3) A is false but R is true

(4) Both A and R are true but R is not the correct explanation of A

Answer (3)

Sol. 

Acid sensitive group also react in clemmensen reduction.

42. Boric acid is solid, whereas BF_3 is gas at room temperature because of

(1) Strong hydrogen bond in Boric acid

(2) Strong covalent bond in BF_3

(3) Strong van der Waal's interaction in Boric acid

(4) Strong ionic bond in Boric acid

Answer (1)

Sol. Boric acid is solid because of strong hydrogen bond in it.

43. Maximum number of electrons that can be accommodated in shell with n = 4 are :

(1) 72 (2) 32

(3) 16 (4) 50

Answer (2)

Sol. Maximum electrons accommodated in (n = 4) is $2n^2 = 32$ electrons

44. Given below are two statements :

Statement I : During Electrolytic refining, the pure metal is made to act as anode and its impure metallic form is used as cathode.

Statement II : During the Hall-Heroult electrolysis process, purified Al_2O_3 is mixed with Na_3AlF_6 to lower the melting point of the mixture.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both Statement I and Statement II are incorrect

(2) Both Statement I and Statement II are correct

(3) Statement I is incorrect but Statement II is correct

(4) Statement I is correct but Statement II is incorrect

Answer (3)

Sol. During electrolytic refining, the pure metal is made to act as cathode and impure metal used as anode.

45. $KMnO_4$ oxidises I^- in acidic and neutral/faintly alkaline solution, respectively, to

(1) I_2 & I_2 (2) I_2 & IO_3^-

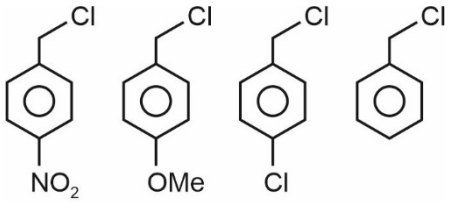
(3) IO_3^- & I_2 (4) IO_3^- & IO_3^-

Answer (2)

Sol. $MnO_4^- + I^- \longrightarrow I_2 + Mn^{2+}$ (acidic medium)

$MnO_4^- + I^- \longrightarrow IO_3^- + MnO_2$ (faintly alkaline)

46. Decreasing order towards S_N1 reaction for the following compounds is

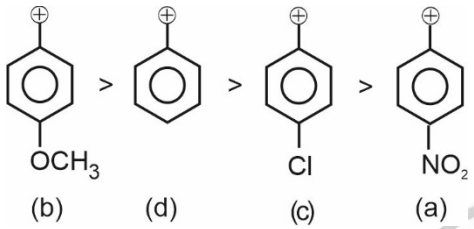


(1) a > c > d > b (2) a > b > c > d

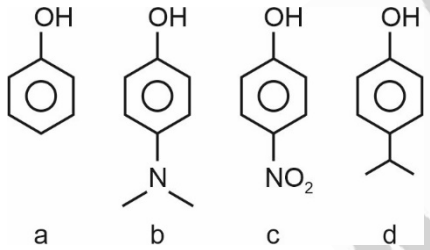
(3) d > b > c > a (4) b > d > c > a

Answer (4)

Sol. Rate of $S_N1 \propto$ stability of carbocation produced



47. The correct order of pK_a values for the following compounds is

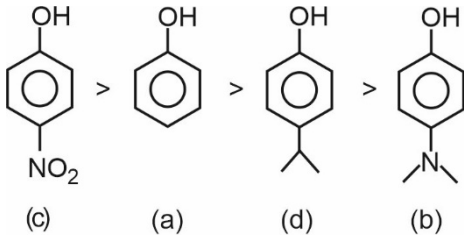


(1) c > a > d > b (2) b > a > d > c

(3) a > b > c > d (4) b > d > a > c

Answer (4)

Sol. The correct acidic order is



$\therefore$ Order of pK_a will be = b > d > a > c

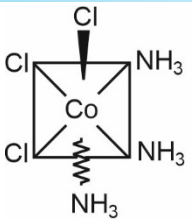
48. The Cl–Co–Cl bond angle values in a fac–$[Co(NH_3)_3Cl_3]$ complex is/are

(1) 180° (2) 90° & 180°

(3) 90° (4) 90° & 120°

Answer (3)

Sol.



fac - $[Co(NH_3)_3Cl_3]$

All the Cl–Co–Cl bond angles are of 90°.

49. The water quality of a pond was analysed and its BOD was found to be 4. The pond has

(1) Slightly polluted water

(2) Very clean water

(3) Highly polluted water

(4) Water has high amount of fluoride compounds

Answer (2)

Sol. A clean water would have BOD value less than 5 ppm.

50. Formulae for Nessler's reagent is

(1) K_2HgI_4 (2) HgI_2

(3) KHg_2I_2 (4) $KHgI_3$

Answer (1)

Sol. Nessler's reagent is K_2HgI_4

SECTION - B

Numerical Value Type Questions: This section contains 10 questions. In Section B, attempt any five questions out of 10. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the second decimal place; e.g. 06.25, 07.00, –00.33, –00.30, 30.27, –27.30) using the mouse andw the on-screen virtual numeric keypad in the place designated to enter the answer.

51. An organic compound undergoes first order decomposition. If the time taken for the 60% decomposition is 540 s, then the time required for 90% decomposition will be is _____s. (Nearest integer).

Given : ln 10 = 2.3; log 2 = 0.3

Answer (1350)

Sol. In 60% decomposition $A_0 = 1$, $A_t = 0.4$

$$k = \frac{1}{t}\ln\left(\frac{A_0}{A_t}\right) = \frac{1}{540}\ln\left(\frac{10}{4}\right)$$

$\therefore$ time t_0 complete 90% reaction

$$t = \frac{1}{k}\ln(10)$$

$$= \frac{\ln(10)}{\ln\left(\frac{10}{4}\right)} \times 540$$

$$= \frac{2.3}{2.3 \times 0.4} \times 540$$

= 1350 s

52. A short peptide on complete hydrolysis produces 3 moles of glycine (G), two moles of leucine (L) and two moles of valine (V) per mole of peptide. The number of peptide linkages in it are _____.

Answer (06.00)

Sol. The peptide has seven amino acid units therefore it has six peptide bonds.

53. 1 mole of ideal gas is allowed to expand reversibly and adiabatically from a temperature of 27°C. The work done is 3 kJ mol^{-1}. The final temperature of the gas is _____K (Nearest integer). Given C_V = 20 J mol^{-1} K^{-1}

Answer (150.00)

Sol. T_1 = 300 K w = 3 kJ/mole

$w = nC_v\Delta T$

$3000 = 1 \times 20 \times (300 - T_2)$

$300 - T_2 = 150$

T_2 = 150 K

54. The electrode potential of the following half cell at 298 K

$X \mid X^{2+}$ (0.001 M) || Y^{2+} (0.01 M) | Y is _____ × 10^{-2} V (Nearest integer).

Given : $E^{\circ}_{X^{2+}|X} = -2.36$ V

$E^{\circ}_{Y^{2+}|Y} = +0.36$ V

$\frac{2.303RT}{F} = 0.06$ V

Answer (275.00)

Sol. $$E_{cell} = E^{\circ}_{cell} - \frac{0.06}{n}\log\frac{x^{2+}}{y^{2+}}$$

$$E_{cell} = (2.36 + 0.36) - \frac{0.06}{2}\log\left(\frac{1}{10}\right)$$

= 2.36 + 0.36 + 0.03

$\simeq$ 2.75 V

55. The graph of $\log\frac{x}{m}$ vs log p for an adsorption process is a straight line inclined at an angle of 45° with intercept equal to 0.6020. The mass of gas adsorbed per unit mass of adsorbent at the pressure of 0.4 atm is _____ × 10^{-1} (Nearest integer).

Given : log 2 = 0.3010

Answer (16.00)

Sol. $\frac{1}{n} = \tan(45°)$ $\therefore n = 1$

$\log k = 0.602$ $\therefore k = 4$

$$\frac{x}{m} = k.p^{1/n}$$

= 4 × 0.4

= 1.6 g

56. Iron oxide FeO, crystallises in a cubic lattice with a unit cell edge length of 5.0 Å. If density of the FeO in the crystal is 4.0 g cm^{-3}, then the number of FeO units present per unit cell is _____. (Nearest integer)

Given : Molar mass of Fe and O is 56 and 16 g mol^{-1} respectively.

$N_A = 6.0 \times 10^{23}$ mol^{-1}

Answer (04.00)

Sol. $d = \frac{z \times m}{a^3}$

$$4 = \frac{z \times 72}{6 \times 10^{23} \left(5 \times 10^{-8}\right)^3}$$

$$4 = \frac{z \times 72}{6 \times 125 \times 10^{-1}}$$

$= z \approx 4$

57. Lead storage battery contains 38% by weight solution of H_2SO_4. The van't Hoff factor is 2.67 at this concentration. The temperature in Kelvin at which the solution in the battery will freeze is _____ (Nearest integer).

Given $K_f = 1.8$ K kg mol^{-1}

Answer (243.00)

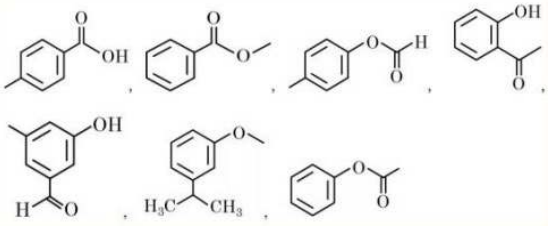
Sol. $\Delta T_f = K_f \, i \, m$

$$= 1.8 \times 2.67 \times \frac{\frac{38}{98}}{0.062}$$

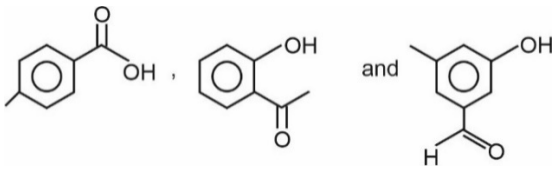
$= 30$

$\therefore$ It freeze at 273 – 30 = 243 K

58. Number of compounds from the following which will not dissolve in cold $NaHCO_3$ and NaOH solutions but will dissolve in hot NaOH solution is _____



Answer (03.00)

Sol.  and

will dissolve in hot NaOH solution

59. The strength of 50 volume solution of hydrogen peroxide is _____ g/L (Nearest integer).

Given :

Molar mass of H_2O_2 is 34 g mol^{-1}

Molar volume of gas at STP = 22.7 L.

Answer (150.00)

Sol. $H_2O_2 \longrightarrow H_2O + \frac{1}{2}O_2$

$$\frac{50}{22.7}$$

$\therefore$ Moles of H_2O_2 in solution = $\frac{50}{22.7} \times 2$

$\therefore$ Strength = $\frac{\frac{50 \times 2}{22.7} \times 34}{1}$

$= 149.78$

≈ 150

60. Consider the following equation:

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g), \quad \Delta H = -190 \text{ kJ}$$

The number of factors which will increase the yield of SO_3 at equilibrium from the following is _____

A. Increasing temperature

B. Increasing pressure

C. Adding more SO_2

D. Adding more O_2

E. Addition of catalyst

Answer (03.00)

Sol. $2SO_2 + O_2 \rightleftharpoons 2SO_3 \quad \Delta H = -190$ kJ

It is an exothermic reaction

$\therefore$ factor B, C, D will increase the amount of SO_3.

MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

61. Let A be a point on the x-axis. Common tangents are drawn from A to the curves $x^2 + y^2 = 8$ and $y^2 = 16x$. If one of these tangents touches the two curves at Q and R, then $(QR)^2$ is equal to

(1) 76 (2) 72

(3) 64 (4) 81

Answer (2)

Sol. Let a tangent on $y^2 = 16x$ be $y = mx + \frac{4}{m}$

For common to $x^2 + y^2 = 8$

$$\frac{4}{m} = 2\sqrt{2}\left(1+m^2\right)$$

$$\Rightarrow \quad \frac{2}{m^2} = 1+m^2 \quad \Rightarrow \quad m = \pm 1$$

Taking one of the tangent $y = x + 4$

Point of tangency with $y^2 = 4x$

$x^2 + 8 + 16 = 4x \quad \Rightarrow x = 4$ & $y = 8$

$\therefore \quad Q(4, 8)$

and for $x^2 + y^2 = 8$

$2x^2 + 8x + 8 = 0$

$x^2 + 4x + 4 = 8 \Rightarrow x = -2, y = 2 \Rightarrow R = (-2, 2)$

$(QR)^2 = 6^2 + 6^2$

$= 72$

62. Let f, g and h be the real valued functions defined of $\mathbb{R}$ as

$$f(x)\begin{cases}\frac{x}{|x|}, & x \neq 0\\ 1, & x = 0\end{cases}, \quad g(x)\begin{cases}\frac{\sin(x+1)}{(x+1)}, & x \neq -1\\ 1, & x = -1\end{cases}$$

and $h(x) = 2[x] - f(x)$, where $[x]$ is the greatest integer $\leq x$.

Then the value of $\lim\limits_{x\to 1} g\left(h\left(x-1\right)\right)$ is

(1) 1 (2) –1

(3) sin(1) (4) 0

Answer (1)

Sol. $f(x) = \text{sgn}(x)$

$h(x) = 2[x] - \text{sgn}(x)$

If $x \to 1^+$ then $h(x-1) = 2[x] - 2 - \text{sgn}\,(x-1)$

$= 0 - 1 = -1$

& if $x \to 1^-$ then $h(x-1) = 2[x] - 2 - \text{sgn}\,(x-1)$

$= -2 + 1 = -1$

$$\therefore \quad \lim_{x\to 1^+} g\left(h\left(x-1\right)\right) = \lim_{x\to 1^+} \frac{\sin\left(h\left(x+1\right)+1\right)}{h\left(x-1\right)+1} = 1$$

$$\lim_{x\to 1^-} g\left(h\left(x-1\right)\right) = \lim_{x\to 1^-} \frac{\sin\left(h\left(x-1\right)\right)+1)}{h\left(x-1\right)+1} = 1$$

63. Let S be the set of all values of a_1 for which the mean deviation about the mean of 100 consecutive positive integers $a_1, a_2, a_3, \ldots\ldots, a_{100}$ is 25. Then S is

(1) {9} (2) {99}

(3) ϕ (4) $\mathbb{N}$

Answer (4)

Sol. Let $a_1 = a \Rightarrow a_2 = a + 1, \ldots\ldots a_{100} = a + 90$

$$\mu = \frac{\frac{100a + (99\times 100)}{2}}{100} = a + \frac{99}{2}$$

$$\text{M.D} = \frac{\sum|a_1 - \mu|}{100} = \frac{\left(\frac{99}{2} + \frac{97}{2} + \ldots\ldots \frac{1}{2}\right)^2}{100} = \frac{2500}{100} = 25$$

$\therefore \quad a \to z$

64. If a plane passes through the points $(-1, k, 0)$, $(2, k, -1)$, $(1, 1, 2)$ and is parallel to the line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$, then the value of $\frac{k^2+1}{(k-1)(k-2)}$ is

(1) $\frac{13}{6}$ (2) $\frac{17}{5}$

(3) $\frac{5}{17}$ (4) $\frac{6}{13}$

Answer (1)